Towards a Tool Supported Approach for Regulatory Requirements Engineering

Parisa Elahidoost
fortiss GmbH, Research Institute of the Free State of Bavaria, Germany
Blekinge Institute of Technology, Sweden
elahidoost@fortiss.org

***Abstract*—With the escalating complexity and range of regulations impacting the development and operations of software-intensive systems, engineers are compelled to manage intensifying regulatory oversight. The critical task of analyzing and interpreting regulatory norms, as well as deriving software requirements, is a vital step in achieving regulatory compliance. Nevertheless, the interpretation of regulations remains heavily reliant on the individual expertise and domain-specific experience of legal professionals, with a notable absence of systematic methodologies and supportive tools to streamline this process. Research in this domain frequently remains isolated from the practical experiences of industry practitioners, resulting in solutions that struggle to find relevance in real-world applications. The work outlines a doctoral thesis aiming to have a detailed examination of the existing state of reported evidence in RE related to regulatory compliance and, analysis of current practices and obstacles in practice, to identify key areas for improvement and development of supportive tools and methodologies. Furthermore, this work includes an investigation into the limitations and potentials of automation in crafting viable approaches for regulatory RE. The ultimate goal is to bridge the theoretical and practical aspects of regulatory RE, ensuring the creation of a tool-supported approach that is both academically robust and pragmatically applicable. By focusing on enhancing the structure and utility of RE practices in the face of regulatory demands, this work seeks to contribute to the field, paving the way for more effective compliance management in software engineering.**

***Index Terms*—Requirements Engineering, Regulatory Compliance, Change-Impact Analysis**

I. Introduction

Software-intensive systems are required to adhere to a wide range of regulatory artifacts, including laws, policies, mandates, and guidelines. In principle, we consider regulations to be official documents containing public, general, and obligatory norms issued by regulatory bodies [2]. The concept of regulatory compliance involves ensuring that an organization, its processes, or its products align with applicable laws, guidelines, specifications, and regulations [1]. Achieving compliance begins with the accurate reflection of these regulatory artifacts within the software requirements specified during the development of software projects. However, this process is prone to potential errors and demands a significant amount of effort [3], often relying heavily on the expertise of legal and domain specialists who may not always be accessible to engineering teams. Additionally, the Information Technology sector is facing escalating regulatory scrutiny on numerous fronts, from privacy concerns to the introduction of cutting-edge technologies [4], requiring software engineers to navigate a complex web of regulations that often apply simultaneously. Insufficient compliance efforts can lead to severe penalties, including substantial fines or the withdrawal of non-compliant products and services from the market. We consider *Regulatory Requirements compliance* as how well functional and non-functional requirements and constraints reflect the criteria specified in regulations. To provide engineering teams with more structured methodologies and supportive tools for elaborating and maintaining requirements in tune with regulatory sources, an initial step involves a thorough understanding of current practices and challenges encountered by these teams. Furthermore, it's crucial to ascertain the current state of reported evidence in Requirements Engineering (RE) concerning regulatory compliance, focusing on identifying challenges, principles, and practices. This knowledge will direct our research and development efforts towards solutions that are problem-driven and practically relevant. The objective of this thesis is to uncover the challenges associated with managing regulatory requirements within RE, exploring the limitations and possibilities of tool support to develop sensible approaches for addressing these challenges. This thesis is rooted in academia-industry collaboration with a medium-sized software development and renovation company, itestra and we plan to focus on one specific domain. The industrial challenge deemed most pressing in the engineering of software-intensive systems for the financial and insurance sector, represented by itestra is the complexity of analyzing variations, like tariffs for different regions or similar union contracts for different businesses, and also change impact analysis in case of updates in regulations. The thesis intends to bridge the gap between the theoretical aspects of regulatory RE and its practical implementation, ensuring that the developed approaches are both theoretically sound and practically viable.

II. Related Work

We conducted a comprehensive review of the literature as part of the systematic mapping study detailed in Section V-A. This section specifically delves into the analysis of recent secondary studies concerning the topic of regulatory compliance.

Ardila et al. [5] present a systematic literature review focused on compliance checking practices within software process development. Published in 2022, this review critically examines the existing body of work to synthesize and evaluate the methodologies and tools that have been proposed to ensure that software processes adhere to relevant regulations. The research identified five overarching categories of challenges pertinent to compliance within the development of software-intensive process systems, yet it did not explicitly associate these categories with particular results from the publications analyzed. Specifically, they mentioned that there is a need to identify appropriate languages that will unify a generic and norm-independent solution, enhance the degree of automation, bolster tool support, and elevate practical application by refining usability features. Mubarkoot et al. [6] conducted a systematic literature review to explore the domain of software compliance requirements, factors, and policies, as elaborated in their 2023 publication. The study rigorously evaluated existing frameworks and standards governing software compliance, pinpointing limitations in current methodologies and highlighting avenues for future research. The research highlights human and technology-related challenges in regulatory compliance, proposing policy-based solutions divided into technological and human-centric categories. Yet, it points out a critical gap in detailed classification crucial for practical application in software engineering compliance. Akhigbe et al. [7] undertook a systematic literature mapping to examine both goal and non-goal modeling methods within the context of legal and regulatory compliance, examining 103 studies. This investigation delved into the advantages and constraints of these methods when applied to compliance-related activities. Specifically, the research spotlighted obstacles associated with goal modeling, pointing out concerns like the constrained utility of goal models which are suited only for comparing models derived from legal documents, rather than evaluating the documents directly. The research conducted by Usman et al. [14], examining compliance requirements in large-scale software development via an industrial case study within a single organization, specifically targeting the encountered challenges and practices. Although it provides essential insights into strategies for managing compliance and identifies specific challenges within the industry, its primary limitation lies in the narrow focus on just one case. A study tcloser with the thesis's specific focus was conducted by Sleimi et al. [13]. This research advanced the systematic traceability of legal requirements through the development of a methodology for the automatic extraction of metadata, utilizing domain-specific semantics.

This research endeavors to bridge the identified gap in understanding how software engineering navigates regulatory compliance requirements. While prior studies have shed light on the various challenges and have suggested numerous compliance verification and management techniques, they tend to either omit an in-depth classification of these challenges or concentrate narrowly on certain compliance facets like modeling techniques or policy implementation. These prior studies lay a foundational understanding of compliance verification and the technological and human elements involved, yet they often do not delve into the nuanced application of these findings within the practical realm of software engineering. This thesis aspires to deliver a holistic insight into the incorporation of regulatory mandates into software development workflows, pinpoint prevailing challenges, and examine opportunities for enhancements and the development of supportive tools.

III. Research Goal

The goal of this research is to understand the possibilities and limitations of automation in regulatory requirements and develop a sensible approach along those limitations to support engineers in dealing with their most pressing challenges. The intent is to develop approaches in RE that not only recognize the inherent challenges but also harness the potential of tool support, thereby refining the process of complying with regulatory mandates in the field of software engineering. The thesis begins with a problem analysis and uses the results for further scoping in an industrial context with a focus on one specific domain. We aim to address the study goal by answering the following research questions:

RQ1 **What is the current state of reported evidence in regulatory RE?**

RQ1.1 What are the reported challenges in regulatory compliance of software intensive systems?

RQ1.2 What is needed to address the challenges, and what principles and practices are used?

Motivation: To outline the current landscape of reported evidence in the field of requirements engineering for the regulatory compliance of software-intensive systems. With a specific focus on the challenges encountered, as well as the principles and practices implemented to overcome these hurdles, the aim is to elaborate a comprehensive overview of existing research. The aim is to pave the way for future scholarly inquiry and practical applications, offering valuable insights to both researchers and practitioners engaged in enhancing regulatory compliance within software engineering.

RQ2 **What are the practices and challenges when handling requirements with regulation as the source in the industry?**

RQ2.1 What are the engineering practices when working with requirements with regulations as a source?

RQ2.2 What are the associated challenges encountered by software engineering teams when working with requirements stemming from regulations?

RQ2.3 Which engineering activities, highlighted by the identified challenges, can benefit most from tool support?

Motivation: Firstly, to shed light on how software engineering practices adapt to regulatory requirements, examining the methods used to interpret and embed these regulations within the software development lifecycle. Secondly, to discern and understand the hurdles practitioners are confronted with while handling requirements that originate from regulatory sources into software engineering routines. Identifying such impediments offers insights into prevalent gaps and issues that emerge during the compliance process. Thirdly, to investigate and articulate the specific engineering practices that are hindered by the challenges to pinpoint where tool support can help these complexities, thereby facilitating more efficient compliance processes. This threefold motivation aims to form an understanding of practices, challenges, and technological potential toward streamlining regulatory compliance in software requirements engineering.

RQ3 **How and to what extent is tool support possible to assist engineers in addressing their most critical challenges?** Motivation: To explore tool support as a solution to the challenges unearthed in regulatory requirements engineering. By investigating how tool support can be applied and measuring the extent of its effectiveness and limitations, the aim is to assess the potential for technological aids to address the obstacles identified. This RQ was initially formulated with the intention of further refinement based on the findings from *RQ1* and *RQ2*.

IV. Methodology

Our research methodology unfolds in three distinct phases. The initial phase tackles research questions *RQ1* and *RQ2* through a comprehensive *problem analysis* that spans both academic literature and industry practices. We began by pinpointing the state of reported evidence in requirements engineering for regulatory compliance, employing a systematic mapping study based on the guidelines set by Kitchenham et al. [8], with a specific focus on identifying the challenges and the diverse principles and practices developed to address these challenges. Next, we engaged with industry professionals to gain a grounded understanding of the practicalities of handling requirements derived from regulations, along with the specific practices and challenges encountered in actual project scenarios. Utilizing a multiple-case study approach as outlined by Runeson et al. [11], we examined four cases, harnessing a trio of methods: semi-structured interviews, group discussions, and analysis of relevant corporate documentation. This approach enriched our data collection, ensuring a robust and comprehensive set of insights to address *RQ2*. The results from these studies are then used to update the *RQ3* and form the direction of the development of a sensible tool-supported approach for addressing the prominent challenge while in regulatory RE. We aim to follow a design science paradigm [10], [9] to design, implement, and explore tool support limitations in the *development* phase of the research. In the third phase of this thesis, *continuous and iterative evaluation*, our plan is to assess the tool-supported approach through a case study within an industrial context. *RQ3* is steered by the principles of technical action research and the development and continuous evaluation will be jointly with our industry partner. Design science is a cyclic approach with several iterations and the V-D and V-E are currently planned.

V. Current and Planned Contributions

A. *Systematic Mapping Study (RQ1)*

Software-intensive systems are required to adhere to regulatory artifacts, including laws, policies, mandates, and guidelines. Compliance is initiated by accurately incorporating these regulatory artifacts into the software requirements formulated at the outset of software project development. It is imperative to have a comprehensive understanding of the current state of research in the area of requirements engineering for regulatory compliance. To this end, we have undertaken a systematic mapping study (SMS) analyzing over 136 studies published within (2017, 2021). Through this SMS, we have identified and categorized the RE-related challenges (11 categories) in regulatory compliance of software-intensive systems and their potential connection to six types of principles and practices addressing challenges. For example, the use of models was prominent in addressing challenges in relation to domain knowledge and expert human resources. Notably, there is a lack of tool support for addressing the identified challenges. Additionally, our analysis revealed that approximately 13% of the primary studies emphasized the role of legal experts in the formulation of principles and practices. The majority of these studies concentrated on a select number of regulatory fields (such as privacy and safety) and application domains (including healthcare and avionics). This SMS is currently undergoing peer review for publication in a journal.

B. *Multi-Case Study (RQ2)*

We conducted an industrial multi-case study with four distinct cases to explore contemporary industrial practices and challenges when inferring requirements from regulations to support more problem-driven research. We collected the data through a triad of methods: conducting interviews, facilitating group discussions, and examining relevant corporate documentation. We observed a variety of approaches to handling regulatory requirements in software development, including the interpretation of regulations and collaboration with domain experts. Furthermore, we pinpointed challenges associated with embedding regulatory requirements into software development, such as the complexity of legal terminology, communication barriers between domain experts and developers, the effects of delayed regulatory updates, and the obstacles in assessing how regulatory changes influence current systems. In joint discussions with the industry partner we pinpointed critical areas that could benefit from tool-supported strategies. Our contribution is a comprehensive case study that informs a more nuanced understanding of the industry's needs, and a set of recommendations for future problem-driven research directions. This study is, at the time of writing this manuscript, under review at the industrial track of a conference.

C. *First Prototype (RQ3)*

Following the design science approach, our investigation delved into the intricacies of analyzing and tracking requirements within the framework of contractual legal documents, with a deliberate emphasis on managing the variability of these requirements which was the area identified as most pressing in the company. Across five iterative phases, we collaboratively crafted, assessed, and refined our solution, bringing to light valuable strategies and insights within the domain of managing requirements variability in the legal and contractual domain. Our ambition was to forge an automated system capable of effectively and precisely recognizing and orchestrating the commonalities across varying contractual documents. An observation through our study was the understanding that achieving full automation in identifying shared configurations might not be completely achievable. Our findings highlight the indispensable role of expert judgment, especially for tasks that necessitate legal or contractual scrutiny. Nevertheless, the stakeholders involved in our study acknowledged that even a modest degree of automation can be beneficial in reducing the reliance on manual processes, minimizing errors, and improving overall efficiency in the management of soft- ware requirements configurations. While these insights are particularly tied to this context, they open up possibilities for applying this approach across various industries to tackle similar challenges. This study has been accepted for presentation at the REFSQ 2024 conference (Requirements Engineering: Foundation for Software Quality) [12], underscoring its contribution towards evolving practices in requirements engineering.

D. Tool Supported Approach - Future Research Plan (RQ3)

The expected outcome of this research project is a tool-supported approach. We elaborated on the identified challenges from academia and industry and updated the *RQ3*. We aim to provide requirements engineers with tool-based capabilities for change impact analysis and traceability, aiding them in navigating the intricate and evolving landscape of regulatory requirements. This phase will delve into the feasibility of automating the analysis of how regulatory changes affect requirements, intending to integrate established practices with innovative solutions to address the challenges identified. Key issues highlighted in the SMS V-A, such as the processing complexities of regulations, their practical implementation challenges, and the existing gaps in automation and tool support, will be tackled. Additionally, the challenge of change impact analysis and the management of variations, which emerged as significant in three out of four industrial case studies V-B, will be addressed. Our approach includes first probing into the nature of regulatory changes, discerning patterns of change, and determining how these changes can be not only detected but also precisely interpreted regarding their nature and importance, ideally via automated means. Second, after the change identification we plan to focus on the support of impact analysis. We will undertake an exploration of the limitations and potential of automation in change impact analysis. Adopting the design science methodology, we aim to iteratively develop and empirically evaluate our solution, starting from an initial solution proposed in V-C.

E. Evaluation of the Approach - Future Research Plan (RQ3)

To assess the efficacy of the proposed tool-supported approach designed to enhance requirements engineers' capabilities in change impact analysis and traceability, we plan to iteratively with development execute evaluation. This evaluation will scrutinize the approach's ability to assist engineers in managing the complexities and dynamism of regulatory requirements. Employing a mixed-methods strategy, we will blend quantitative and qualitative research techniques to gather a comprehensive data set. Initially, quantitative data will be collected via surveys and questionnaires targeted at requirements engineering professionals, focusing on evaluating the approach's efficiency and the degree to which it addresses the identified challenges. Following this, qualitative data will be acquired through detailed interviews with practitioners to derive deeper insights into the solution's practicality, usefulness, and capacity to automate the interpretation of regulatory changes accurately. The results of these investigations will be the input for the followed by further development. This analysis will not only shed light on the automation's boundaries and capabilities in the context of change impact analysis but also inform continuous improvement of the tool-supported approach. The outcomes of this evaluation will enrich the literature on tool efficacy and support the advancement of methodologies for incorporating change impact analysis and traceability in regulatory RE.

VI. Work Plan and Schedule

Following is the proposed timeline for the Ph.D. research activities:

- **Year 1 & 2:** Identification and analysis of state-of-the-art in requirements engineering for regulatory compliance of software intensive systems with a particular focus on challenges and utilized principles and practices. (Completed) Exploring software engineering practices and identifying challenges when working with requirements that emerge from regulatory sources by practitioners via a case study. (Completed)
- **Year 3 & following** Development of a sensible tool-supported approach to tackle the major challenges identified in regulatory RE, including an investigation into its limitations. Conducting an empirical evaluation of the approach through a case study in an industrial setting, followed by the composition of the thesis. (In progress)

VII. Conclusion

This research makes contributions to the field of regulatory RE by first exploring the challenges addressed in research and within the industry and the approaches used to address them. It then critically analyzes the feasibility of deploying tool support to tackle these identified challenges, exploring the boundaries of current technology and methods. The cornerstone of this research is the development and subsequent evaluation of an innovative tool-supported approach, designed to address the most prominent identified challenge unearthed in regulatory RE. This not only sets a new standard in the practice but also offers empirical evidence on the practicality and efficiency of tool-supported solutions in improving compliance processes within software engineering.

REFERENCES

[1] Akhigbe, O., Amyot, D., Richards, G.: A systematic literature mapping of goal and non-goal modelling methods for legal and regulatory compliance. Requirements Engineering 24(4), 2019.

[2] R. Brownsword, H. Somsen, Law, innovation and technology: before we fast forward—a forum for debate, Law, Innovation and Technology 1 (2009).

[3] M. Kellogg, M. Schaf, S. Tasiran, M. D. Ernst, Continuous compliance, in: 2020 35th IEEE/ACM International Conference on Automated Software Engineering (ASE), IEEE, 2020.

[4] D. Hays, Shaping the future of tech industry regulation 5 steps to take now, 2019.

[5] J. P. Castellanos Ardila, B. Gallina, F. Ul Muram, Compliance checking of software processes: A systematic literature review, Journal of Software: Evolution and Process 34 (2022).

[6] M. Mubarkoot, J. Altmann, M. Rasti-Barzoki, B. Egger, H. Lee, Software compliance requirements, factors, and policies: A systematic literature review, Computers & Security (2023)

[7] O. Akhigbe, D. Amyot, G. Richards, A systematic literature mapping of goal and non-goal modelling methods for legal and regulatory compliance, Requirements Engineering 24,(2019).

[8] B. Kitchenham, S. Charters, Guidelines for performing systematic literature reviews in software engineering, 2007.

[9] Runeson, P., Engström, E., Storey, M.A.: The Design Science Paradigm as a Frame for Empirical Software Engineering, pp. 127–147. Springer (2020).

[10] R. Wieringa, Design Science Methodology for Information Systems and Software Engineering. Springer (2014).

[11] P. Runeson and M. Höst, "Guidelines for conducting and reporting case study research in software engineering" Empirical Software Engineering, 14, 131-164, 2009.

[12] P. Elahidoost, M. Unterkalmsteiner, D. Fucci, P. Liljenberg and J. Fischbach, Designing NLP-based solutions for requirements variability management: experiences from a design science study at Visma, 2402.07145, arXiv, 2024.

[13] Sleimi, A., Sannier, N., Sabetzadeh, M., Briand, L., Dann, J.: Automated extraction of semantic legal metadata using natural language processing. In: 26th International Requirements Engineering Conference (RE). IEEE (2018)

[14] Usman, M., Felderer, M., Unterkalmsteiner, M., Klotins, E., Mendez, D., Alégroth, E. (2020). "Compliance Requirements in Large-Scale Software Development: An Industrial Case Study." In: Morisio, M., Torchiano, M., Jedlitschka, A. (eds) Product-Focused Software Process Improvement. PROFES 2020, vol 12562. Springer, Cham.